

UNIVERSIDAD PRIVADA DE TACNA
FACULTAD DE INGENIERIA

Escuela Profesional de Ingeniería de Sistemas



Avance del proyecto

Curso:

Machine Learning

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1. Descripción del problema (clasificación).

DATASET SELECCIONADO: Bank Marketing

<https://www.kaggle.com/datasets/henriqueyamahata/bank-marketing>

```
import sys
assert sys.version_info >= (3, 7)

from packaging import version
import sklearn
assert version.parse(sklearn.__version__) >= version.parse("1.0.1")

from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

np.random.seed(42)

# — Carga del dataset (Google Colab) —
def load_bank_data():
    csv_path = Path("/content/sample_data/bank-additional-full.csv")
    return pd.read_csv(csv_path, sep=";")

bank = load_bank_data()

# — Exploración rápida —
print(bank.shape)
print(bank.head())
print(bank["y"].value_counts())

# — Split estratificado por la variable objetivo —
from sklearn.model_selection import train_test_split

bank["y_cat"] = (bank["y"] == "yes").astype(int)

strat_train_set, strat_test_set = train_test_split(
    bank, test_size=0.2, stratify=bank["y_cat"], random_state=43
)

for set_ in (strat_train_set, strat_test_set):
    set_.drop("y_cat", axis=1, inplace=True)

bank = strat_train_set.copy()
```

```

for set_ in (strat_train_set, strat_test_set):
    set_.drop("y_cat", axis=1, inplace=True)

bank = strat_train_set.copy()

print(f"\nTrain: {len(strat_train_set)} filas | Test: {len(strat_test_set)} filas")
print("Proporción 'yes' en train:", (strat_train_set["y"] == "yes").mean().round(4))
print("Proporción 'yes' en test: ", (strat_test_set["y"] == "yes").mean().round(4))

```

```

... (41188, 21)
   age  job marital education default housing loan contact \
0  56  housemaid married basic.4y no no no telephone
1  57  services married high.school unknown no no telephone
2  37  services married high.school no yes no telephone
3  40  admin. married basic.6y no no no telephone
4  56  services married high.school no no yes telephone

   month day_of_week ... campaign pdays previous poutcome emp.var.rate \
0  may mon ... 1 999 0 nonexistent 1.1
1  may mon ... 1 999 0 nonexistent 1.1
2  may mon ... 1 999 0 nonexistent 1.1
3  may mon ... 1 999 0 nonexistent 1.1
4  may mon ... 1 999 0 nonexistent 1.1

   cons.price.idx cons.conf.idx euribor3m nr.employed y
0  93.994 -36.4 4.857 5191.0 no
1  93.994 -36.4 4.857 5191.0 no
2  93.994 -36.4 4.857 5191.0 no
3  93.994 -36.4 4.857 5191.0 no
4  93.994 -36.4 4.857 5191.0 no

[5 rows x 21 columns]
y
no 36548
yes 4640
Name: count, dtype: int64

Train: 32950 filas | Test: 8238 filas
Proporción 'yes' en train: 0.1127
Proporción 'yes' en test: 0.1126

```

2. Descripción del conjunto de datos (info, describe, value_counts(), hist).

```
print("=" * 60)
print("ANÁLISIS EXPLORATORIO DE DATOS")
print("=" * 60)

print("\n--- Primeras 5 filas del dataset ---")
print(bank.head())

print("\n--- Información general del dataset ---")
bank.info()
# Observación: 'total_bedrooms' tiene menos de 16512 entradas → hay valores NaN.
# 'ocean_proximity' es la única variable categórica; el resto son numéricas.

print("\n--- Estadísticas descriptivas ---")
print(bank.describe())
# Observación: 'median_income' parece estar en decenas de miles (no en dólares directos).
# 'housing_median_age' y 'median_house_value' muestran valores truncados en su máximo,
# lo que indica que los datos fueron recortados artificialmente (ej. cap en $500,000).
```

*** =====

ANÁLISIS EXPLORATORIO DE DATOS

=====

--- Primeras 5 filas del dataset ---

	age	job	marital	education	default	housing	loan	\
29880	46	services	married	high.school	no	no	no	
33055	24	services	single	high.school	no	yes	no	
2961	31	admin.	single	university.degree	no	no	no	
16152	58	retired	married	basic.4y	no	no	no	
28134	26	admin.	married	high.school	no	yes	no	

	contact	month	day_of_week	...	campaign	pdays	previous	\
29880	cellular	apr	mon	...	4	999	0	
33055	cellular	may	mon	...	3	999	0	
2961	telephone	may	wed	...	3	999	0	
16152	cellular	jul	tue	...	1	999	0	
28134	cellular	apr	mon	...	10	999	0	

	poutcome	emp.var.rate	cons.price.idx	cons.conf.idx	euribor3m	\
29880	nonexistent	-1.8	93.075	-47.1	1.405	
33055	nonexistent	-1.8	92.893	-46.2	1.299	
2961	nonexistent	1.1	93.994	-36.4	4.859	

```

...      poutcome emp.var.rate  cons.price.idx  cons.conf.idx  euribor3m  \
29880  nonexistent      -1.8         93.075         -47.1         1.405
33055  nonexistent      -1.8         92.893         -46.2         1.299
2961   nonexistent      1.1         93.994         -36.4         4.859
16152  nonexistent      1.4         93.918         -42.7         4.961
28134  nonexistent      -1.8         93.075         -47.1         1.466

```

```

      nr.employed  y
29880      5099.1  no
33055      5099.1  no
2961       5191.0  no
16152      5228.1  no
28134      5099.1  no

```

[5 rows x 21 columns]

--- Información general del dataset ---

<class 'pandas.core.frame.DataFrame'>

Index: 32950 entries, 29880 to 25815

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	age	32950 non-null	int64
1	job	32950 non-null	object
2	marital	32950 non-null	object
3	education	32950 non-null	object
4	default	32950 non-null	object
5	housing	32950 non-null	object
6	loan	32950 non-null	object
7	contact	32950 non-null	object
8	month	32950 non-null	object
9	day_of_week	32950 non-null	object
10	duration	32950 non-null	int64
11	campaign	32950 non-null	int64
12	pdays	32950 non-null	int64
13	previous	32950 non-null	int64
14	poutcome	32950 non-null	object
15	emp.var.rate	32950 non-null	float64
16	cons.price.idx	32950 non-null	float64